

BioSentinel 2.0

AI Bioreactor Digital Twin, Fault Diagnosis & Counterfactual Prescriptive Control

1. Executive Summary & Judge Pitch Brief

BioSentinel 2.0 is an industrial-grade digital twin that fuses real-time mechanistic bioprocess physics (dual-Monod differential equations) with state-of-the-art machine learning (LightGBM 184-feature classifier, **Macro F1 = 94.6%**) and a dynamic counterfactual rollout engine. The platform forecasts reactor trajectories, detects microbial and mechanical anomalies hours before irreversible damage occurs, and computes ranked operator interventions.

Unlike black-box monitoring tools, BioSentinel explicitly tracks **Recoverability R(t)** and the **Dynamic Intervention Window (DIW)**, providing plant operators with an exact countdown to the **Point of No Return (PNR)**.

2. Reviewer Revisions & Integrity Audit (Applied Immediately)

All three critical feedback items identified during pre-pitch evaluation have been audited and implemented:

Issue Identified	Root Cause	Implementation & Resolution	Status
1. Plot Glitch: Diagonal Loopback	Unsorted telemetry timestamps and duplicate time steps caused Plotly to connect curves back to (0,0).	Applied strict sorting and deduplication: df = df.sort_values(by='timestamp').drop_duplicates(subset=['timestamp']) Added monotonic timestamp guard in chart extension logic.	RESOLVED
2. Contamination Metric Contradiction	Contamination cannot be cured by physical actuators (RPM/Feed). Reporting 19.8% recovery and 185 min DIW contradicted bioprocess reality.	Set Recovery R(t) = 0.0% (IRREVERSIBLE) and DIW = 0 min (BREACH DETECTED) post-onset. Replaced actuator advice with TRIGGER CIP / HARVEST ABORT . Preserved active recovery (>80%) for mechanical faults.	RESOLVED
3. Banner Accent Color Sync	Status dot and progress underline remained neon green even when active fault states (Contamination / kLa / Overfeed) were triggered.	Synchronized dot and accent line to dynamically switch to Amber (#f59e0b) for kLa/Overfeeding and Crimson (#ef4444) for Contamination. Neon green (#22c55e) reserved strictly for Healthy baseline.	RESOLVED

3. Real-Time Telemetry & Metric Ribbon Architecture

The live cockpit delivers immediate visibility into physical sensors and AI-derived outputs side-by-side:

DISSOLVED 0■	RQ (CER / OUR)	OUR (Transfer)	RECOVERY R(t)	DIW (Countdown)	BIOMASS X
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DO concentration (mg/L). Critical threshold: 2.2 mg/L.	Respiratory Quotient. Nominal ~1.0; >1.2 indicates overflow or contamination.	Oxygen Uptake Rate g/(L·h). Reflects true metabolic respiration rate.	Probability of biological recovery upon intervention (0–100%).	Dynamic Intervention Window in minutes until PNR threshold.	Cell dry weight concentration (g/L), verified by OD600.
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4. Bioreactor Operational Scenarios & Fault Response

BioSentinel monitors 4 distinct bioprocess regimes simulated via 4th/5th-order Runge-Kutta ODEs:

Regime Scenario	Onset & Signatures	AI / Physics Output	Prescribed Action
■ Healthy (Baseline)	Nominal dual-Monod growth; DO sag compensated by nominal agitation; RQ ~1.0.	R(t) = 100.0% DIW: Safe (>99 h) Accent: Neon Green (#22c55e)	Nominal control; maintain baseline feed and 400 RPM.
■■ kLa Limitation	Fault onset at t = 8.0 h . Sparger fouling or viscosity rise drops kLa by 70%. DO sags below 2.0 mg/L.	R(t) = 80.2% (at 18h) DIW: ~185 min Accent: Amber (#f59e0b)	+150 RPM Agitation (boosts $kLa \propto RPM^{0.6}$) restores DO above critical limit.
■ Substrate Overfeed	Fault onset at t = 6.0 h . Feed pump over-delivers substrate (3.0x nominal). Causes Crabtree overflow metabolism; RQ spikes.	R(t) = 84.7% DIW: ~140 min Accent: Amber/Orange (#f97316)	-30% Feed Rate Trim immediately curbs substrate accumulation and restores aerobic respiration.
■■ Contamination	Fault onset at t = 5.0 h . Secondary contaminant population (X_c) enters batch, outcompeting culture and skewing RQ.	R(t) = 0.0% (IRREVERSIBLE) DIW: 0 min (BREACH DETECTED) Accent: Crimson (#ef4444)	TRIGGER CIP / HARVEST ABORT. Actuation (+RPM/-Feed) ineffective; prevent cross-contamination.

5. Mathematical Formulation & Kinetic Equations

A. Dual-Monod Specific Growth Rate:

$$\mu(S, C_L) = \mu_{\max} \cdot \left[\frac{S}{(K_S + S)} \right] \cdot \left[\frac{C_L}{(K_O + C_L)} \right]$$

where $\mu_{\max} = 0.40 \text{ h}^{-1}$, $K_S = 0.10 \text{ g/L}$ (substrate affinity constant), and $K_O = 0.0005 \text{ g/L}$ (oxygen affinity constant).

B. Mass Balance Differential Equations:

$$dX/dt = (\mu - D) \cdot X$$

$$dS/dt = D \cdot (S_{\text{feed}} - S) - (\mu / Y_{XS} + m_S) \cdot X$$

$$dC_L/dt = kLa \cdot (C_L^* - C_L) - OUR$$

where $D = F / V$ is dilution rate, C_L^* is saturation dissolved oxygen (0.008 g/L at 1 atm, 37°C).

C. Transfer Rates & Respiratory Quotient:

$$OUR = (\mu / Y_{XO} + m_O) \cdot X \quad | \quad CER = (\mu / Y_{XC} + m_C) \cdot X \quad | \quad RQ = CER / OUR$$

D. Dynamic Stress Accumulation & Recoverability:

Stress $\lambda(t)$ accumulates strictly post fault-onset based on biophysical deviations:

- **kLa Limitation:** $\lambda(t) = 0.25 \cdot \max(2.2 - DO_mg, 0.0)$
- **Substrate Overfeeding:** $\lambda(t) = 0.06 \cdot \max(S - 7.5, 0.0)$
- **Recoverability:** $R(t) = \exp(- \int_{t_{onset}}^t \lambda(\tau) \, d\tau)$
- **Dynamic Intervention Window:** $DIW = [-\ln(R_{pnr}) - \int_{t_{onset}}^t \lambda(\tau) \, d\tau] / \lambda(t)$

6. Machine Learning Classifier Benchmarks

The model is a 4-class LightGBM gradient boosted decision tree trained on 184 rolling telemetry features (windows: 5 min, 15 min, 30 min, 60 min). Pre-onset label denoising was applied to eliminate premature false alarms.

Class Label	Precision	Recall	F1-Score	Support (pts)
Healthy	99.9%	99.9%	99.9%	28,100
kLa Limitation	91.3%	91.9%	91.6%	2,810
Substrate Overfeeding	87.9%	93.0%	90.4%	2,810
Contamination	99.4%	93.9%	96.5%	2,810
Macro Average	94.6%	94.7%	94.6%	36,530

Pitch Ready Certification: BioSentinel 2.0 has been fully verified against biological and mathematical constraints. All telemetry plots are guaranteed monotonic without loopbacks, contamination recovery is appropriately marked irreversible, and banner color-coding is synchronized with active fault states.